PAPER ID-410696

Subject Code: KCE061

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VI) THEORY EXAMINATION 2023-24
ADVANCE STRUCTURAL ANALYSIS

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

a.	What are statically indeterminate structures.	2
b.	Differentiate between Pin-jointed & Rigid-jointed structures.	2
c.	What is meant by influence lines?	2
d.	What will be the absolute maximum bending moment in a simply supported beam when two concentrated wheel loads cross it?	2
e.	What is the use of a suspension bridge?	2
f.	How does suspension bridge work?	2
g.	Write the element stiffness matrix for a truss element.	2
h.		2
i.	What is difference between plastic hinge and mechanical hinge?	2
j.	List out the assumptions made for plastic analysis.	2

SECTION B

2. Attempt any *three* of the following:

a.	A continuous beam ABCD 16 m long is simply supported at A, B, C, D. the beam consists of spans AB, BC, & CD of lengths 4 m, 6m, & 6m respectively. It carries a point load of 64 KN on the span "AB" at a distance of 1m from "A", a load of 45 KN on the span "BC" at a distance of 2 m from "B" and a UDL of 12 KN/m on the span "CD". Find the support moments & reactions.	10
b.	Explain in detail about the necessity of Influence lines and elaborate its applications.	10
c.	Explain the effect of temperature on the horizontal thrust of a two hinged arch subjected to a system of vertical loads?	10
d.	The three hinged stiffening girder of a suspension bridge of span 120m is subjected to two-point loads of 240 KN & 300 KN at distances 25 m & 80 m from the left end. Find the shear force & bending moment for the girder at a distance of 40 m from the left end. The supporting cable has a central dip of 12 m. Find also max. tension in the cable.	10
e.	Calculate the shape factor for a (a)rectangle section of breadth'b' and depth 'd', (b)diamond section of breadth'b' and depth'd'.	10

SECTION C

3. Attempt any *one* part of the following:

a.	A fixed beam of length 6 m carries two-point loads of 30 kN each at a distance of 2 m from both ends. Determine the fixed end moments.	10
b.	Give the procedure for analyzing the continuous beams with fixed ends using three moment equations?	10

4. Attempt any *one* part of the following:

a.	Draw the ILD for shear force and bending moment for a section at50m from the left-hand support of a simply supported beam, 200m long, hence calculate the	10

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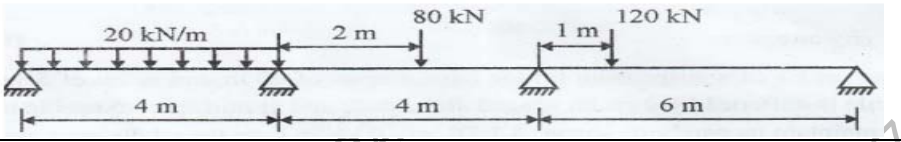
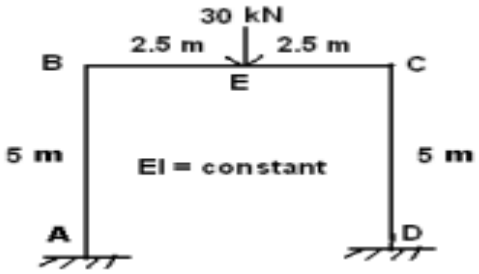
TIME: 3 HRS **M.MARKS: 100**

	maximum bending moment and shear force at the section due to an uniformly distributed rolling load of length 80m and intensity 100kN/m.	
b.	Draw the influence line for the continuous beam ABC simply supported at A and C using Muller Breslau`s principle. AB=3m, BC=4m.EI is constant.	10

5. Attempt any *one* part of the following:

a.	A single load of 100 KN rolls along a girder of 20 m span. Draw the diagrams of maximum bending moment and shear force (positive and negative). What will be the absolute maximum positive shear force and bending moment?	10
b.	A three hinged stiffening girder of a suspension bridge of 100 m span subjected to two-point loads 10 kN each placed at 20 m and 40 m, respectively from the left-hand hinge. Determine the bending moment and shear force in the girder at section 30 m from each end. Also determine the maximum tension in the cable which has a central dip of 10 m.	10

6. Attempt any *one* part of the following:

a.	Analyze the Continuous beam shown in Fig. using flexibility method and draw Bending diagram. 	10
b.	Analyse the portal frame ABCD shown in figure by stiffness method and also draw the bending moment diagram. 	10

7. Attempt any *one* part of the following:

a.	Find the shape factor of the I-section with top flange 100mm wide, bottom flange 150mm wide,20mm tk and web depth 150mm and web thickness 20mm	10
b.	Determine the collapse load for the frame shown in the diagram, Mp is the same for all members.	10

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